

Korea's Nuclear and Energy Renaissance

APR-1400, SMRs and the Carbon-Free Export Machine

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Executive Summary

South Korea has achieved what no other country has managed in the post-Fukushima era: a genuine nuclear renaissance, combining the restart of previously deferred reactor projects, new-build approvals, and an expanding export programme that positions Korea as the world's most commercially competitive nuclear power developer outside of Russia and China. The Czech Republic's selection of Korea Hydro & Nuclear Power (KHNP) for the Dukovany expansion project — worth approximately \$17 billion, the largest nuclear contract in Central European history — confirmed Korea as a credible alternative to the Western nuclear industry's two remaining viable exporters (EDF of France and Westinghouse of the US).

Korea's domestic nuclear programme, which the previous Moon Jae-in government (2017-2022) had placed on a phase-out trajectory, has been comprehensively reversed under President Yoon Suk-yeol (2022-present). New nuclear construction approvals, life-extension decisions for operating reactors, and a target of 30+ nuclear share of electricity generation by 2030 represent a policy reversal unprecedented in scale among major democracies. Meanwhile, KHNP's i-SMR (innovative Small Modular Reactor, 170 MWe) programme targets first commercial operation by 2035, positioning Korea to compete in the emerging SMR market where Western companies (Rolls-Royce, TerraPower, NuScale) are dominant in narrative if not yet in deployed capacity.

Part I — Korea's Nuclear Fleet: The World's Most Operationally Efficient

26 Reactors, 57% Capacity Factor Recovery

Korea operates 26 nuclear power reactors with a combined installed capacity of approximately 24.7 GWe, providing approximately 30-32% of Korea's total electricity generation (the target is to increase this to 35%+ by 2030). The fleet consists predominantly of OPR-1000 (1,000 MWe pressurised water reactors, 18 units) and APR-1400 (1,400 MWe advanced PWR, 6 units in Korea, 4 under construction).

Korea's nuclear fleet has one of the highest capacity factors in the world — consistently above 80% and targeting above 85% under the current government's operational excellence programme. The fleet's performance is driven by Korea's unusually disciplined nuclear operations culture: KHNP operates 26 reactors from a single national utility, enabling standardised maintenance practices, shared spare parts inventories, cross-trained workforce deployment, and systematic lessons-learned implementation that multi-owner, multi-site national programmes cannot replicate.

APR-1400: The Export Champion

The APR-1400 (Advanced Power Reactor 1,400 MWe) is Korea's export flagship — a Generation III+ pressurised water reactor with passive safety features, 60-year design life, and construction cost

credentials that Western developers envy. The APR-1400's commercial credentials were established in the UAE:

Barakah Nuclear Power Plant (UAE)

Unit 1: Commercial operation May 2021

Unit 2: Commercial operation March 2023

Unit 3: Commercial operation October 2023

Unit 4: Commercial operation May 2024

Total capacity: 5,600 MWe (four APR-1400 units)

The UAE Barakah project delivered all four units with capital cost of approximately \$24-27B for the complete 4-unit plant — approximately \$5,500/kWe. This is significantly below the most recent European nuclear projects (Hinkley Point C in the UK: projected \$40B+ for 3.2 GWe = ~\$12,500/kWe) and competitive with the historical Chinese nuclear construction cost benchmark (~\$3,000-4,000/kWe for ACPR-1000 domestic projects).

The UAE project's construction timeline — first unit achieved first criticality on schedule, subsequent units delivered ahead of revised schedule — is Exhibit A in KHNP's export pitch. Korean nuclear construction competence is not theoretical; it is demonstrated at commercial scale in a desert environment, under international oversight, by a non-domestic utility client.

Part II — The Czech Republic Deal: Europe's Most Consequential Nuclear Contract

Dukovany: A \$17 Billion Vindication

The Czech Republic's selection of KHNP as preferred contractor for two new APR-1400 units at the Dukovany site (in the South Moravian region, where the existing 4-unit Soviet VVER-440 plant is located) represents the most consequential nuclear procurement decision in Central Europe in a generation. The contract, signed in principle in July 2024 and finalized in 2025, has a base value of approximately \$17-20 billion (CZK 400-470 billion), with additional scope for Temelin (potential 2 more units) taking the total potential relationship to \$40B+.

The Czech selection process was explicitly geopolitical as well as commercial. The initial tender shortlist (EDF/Framatome, Westinghouse, and KHNP) excluded Russian Rosatom — the operator of the existing Czech nuclear fleet — following Russia's invasion of Ukraine in 2022. The shortlist choice was thus among three Western-allied suppliers with different technical, commercial, and geopolitical profiles.

KHNP's winning proposition included:

- Unit price approximately 10-15% below EDF's Framatome EPR offer (based on Czech government cost assessments)
- Guaranteed construction schedule tied to financial penalties — a commitment that EDF, given Flamanville and Hinkley delays, struggled to replicate
- Technology transfer provisions allowing Czech participation in manufacturing
- US-backed financing (ExIm Bank) to provide government-supported project finance
- US government political support for the KHNP bid, reflecting the US view that Korean nuclear is a "trusted" alternative to Chinese/Russian nuclear in Europe

The Czech deal is transformational for KHNP's commercial credibility. Post-Dukovany, KHNP has active discussions with Poland (potential 3 units at Choczewo and Płótnów sites), Sweden (SAF+ programme),

Netherlands (Borssele 2 expansion), and the UK (Wylfa, though the UK's nuclear procurement is complex).

Part III — The i-SMR: Korea's Next-Generation Export Bet

Why Small Modular Reactors Matter

The SMR market represents nuclear power's attempt to address the fundamental problem that has hobbled large nuclear development in the West: first-of-a-kind construction risk. A 1,400 MWe plant requires 10-15 years of construction and \$15-25B of capital — a risk profile that no utility can absorb without extensive government backstop. SMRs (typically defined as 300 MWe and below) target factory manufacture, standardised module assembly, and shorter construction timelines that reduce first-of-a-kind risk to manageable levels.

Korea's i-SMR (innovative Small Modular Reactor, 170 MWe) is a pressurised water reactor with all primary loop components located inside the reactor pressure vessel (integral design), passive decay heat removal systems requiring no active intervention, and a 60-year design life. The i-SMR programme, funded by the Korean government and KHNP, targets standard design certification from Korea's Nuclear Safety and Security Commission (NSSC) by 2028, with first commercial operation by 2035.

The \$15B annual nuclear export target by 2030 that Korea's government has set requires SMR success alongside APR-1400 large plant exports. Emerging markets — Southeast Asia (Vietnam, Philippines, Indonesia have all expressed interest in nuclear), Africa (Kenya, Ghana, Morocco), and Eastern Europe — have grid systems too small to absorb a 1,400 MWe unit but are potentially well-suited to one or more 170 MWe i-SMR modules. The ASEAN market for i-SMR could be substantial: Indonesia alone is targeting 8,000 MW of nuclear by 2040.

The Competition: Rolls-Royce, NuScale, TerraPower

KHNP's i-SMR faces serious competition in the international SMR market:

Rolls-Royce SMR (470 MWe): The UK government has backed Rolls-Royce's SMR development with £210M in initial funding; the design targets 2030 deployment with UK grid regulatory approval. Rolls-Royce's advantage is brand recognition and the UK's nuclear regulatory reputation; its disadvantage is that no unit has yet been built and the 470 MWe size is larger than optimal for the smallest grids.

NuScale (77 MWe modules, scalable): The US company that pioneered the NRC design certification process for SMRs. NuScale's VOYGR plant received NRC standard design approval in 2022 — the first SMR to do so. However, NuScale's carbon-free power project in Idaho (12-module, 924 MWe total) was cancelled in November 2023 due to cost escalation and customer withdrawal, severely damaging NuScale's commercial credibility. The company's financial viability remains uncertain as of 2026.

CANDU Energy / ARC Clean (fast spectrum): Canadian SMR variants targeting different technical niches, less commercially advanced.

Russia's RITM-200 (50 MWe): Rosatom's floating nuclear power plant variant, operating in Pevek Russia, and the basis for export proposals to developing countries — politically constrained post-Ukraine but technically proven.

Korea's i-SMR enters this competitive landscape with KHNP's operational credibility from 26 domestic reactors and the Barakah international project, plus the commercial backing of Korea's government export promotion machinery. The most significant differentiator relative to Western SMR competition is Korea's demonstrated ability to actually build nuclear power plants on schedule and budget.

Part IV — Domestic Energy Policy: The Yoon Reversal

From Phasout to Renaissance

The policy whiplash in Korean nuclear energy between 2017 and 2022 has few parallels in energy policy history. President Moon Jae-in's 2017 post-Fukushima-inspired nuclear phase-out plan stopped construction of two new APR-1400 reactors (Shin-Hanul 3 and 4), approved no new nuclear construction, and set a formal schedule to close operating reactors as they reached their 40-year design life. Moon's phase-out plan would have reduced nuclear's share of Korean electricity to below 25% by 2030 and near-zero by 2060.

President Yoon Suk-yeol's energy policy (announced in 2022 and confirmed in the 10th Basic Plan for Electricity Supply and Demand 2023) reversed every major element of the Moon plan:

- Shin-Hanul 3 and 4 construction approved and restarted (target operation 2030 and 2031)
- Three new nuclear units approved for construction at Shin-Hanul and Uljin sites (2024-2025 decisions)
- Life extensions approved for existing units: Wolsong 1 (reversed retirement to extended operation), Kori 2 and 3 life-extension decisions
- Nuclear share target raised to 30%+ by 2030, 35%+ by 2036

The policy reversal reflects several intersecting imperatives: Korea's electricity demand is growing, driven by AI data centres and EV charging infrastructure; natural gas imports remain expensive and geopolitically vulnerable following the Ukraine-induced energy price spike; and renewable intermittency (Korea has limited geography for large-scale onshore solar or wind) makes firm baseload power increasingly valuable.

The Coal Phase-Out and Nuclear Baseload

Korea has committed to phase out all coal-fired power generation by 2038 — a commitment that makes nuclear's baseload role even more critical. Korea's electricity system has approximately 36 GWe of coal capacity as of 2025; replacing this capacity with renewables alone would require an impractical density of offshore wind and solar given Korea's geographic constraints. The government's energy mix targets assume nuclear fills the baseload gap left by coal retirement.

Part V — Strategic Outlook: The \$15 Billion Export Target

Export Pipeline Assessment

Korea's nuclear export pipeline as of mid-2026:

Contracted/In Negotiation:

- Czech Republic Dukovany: \$17-20B (contracted); Temelin expansion: under discussion
- Poland (Choczewo site): \$25-30B potential (3 APR-1400 units, active competition with US, France)
- Netherlands Borssele 2: ~\$10B potential (1-2 units, KHNP in final stages)

Active Discussions:

- Sweden: Several utilities evaluating KHNP for new units post-Ringhals revival decision
- UAE additional units: Potential expansion of Barakah site
- Vietnam: Nuclear programme restarted after 2016 suspension; KHNP is the leading candidate
- Philippines: Bataan nuclear plant rehabilitation discussions (KHNP in discussions)

Emerging Markets:

- South Africa (Koeberg expansion): KHNP competing with EDF and Russian/Chinese competitors
- Kenya, Ghana, Morocco: i-SMR potential as grids develop
- Indonesia: Nuclear law recently updated; KHNP is the most actively engaged foreign supplier

The \$15B annual export target requires Korea to win major European contracts (Poland is the critical near-term test) and maintain Dukovany execution excellence. Poland's decision, expected 2026-2027, is the single most important near-term test of KHNP's export momentum. If KHNP wins Poland, the European nuclear supply chain (French and German pressure to buy European) will face a decisive challenge from Korea's commercial proposition. If Westinghouse's AP1000 wins Poland — supported by the US political relationship and Westinghouse's European credential claims — KHNP's European ambition is significantly constrained.

Conclusion: The Nuclear Power That Forgot to Phase Out

Korea's nuclear industry has produced the most consequential energy policy reversal of the 2020s — and the most commercially compelling nuclear export offering in the world outside of Russian Rosatom (which is geopolitically constrained post-Ukraine) and Chinese CGNPC (constrained by nuclear security concerns in Western markets). The APR-1400's UAE commercial track record, KHNP's operational excellence with 26 domestic reactors, and the Czech Dukovany win have created the most credible non-Russian, non-Chinese nuclear export franchise in history.

The i-SMR programme represents the technology bet required to access markets too small for large plants. Its 2035 first-operation target is achievable given Korea's nuclear regulatory experience, though not guaranteed. The \$15B export target by 2030 requires winning at least one major European project (Poland is the decisive test) and maintaining current Dukovany execution.

Korea's nuclear renaissance is simultaneously a domestic energy security story (reducing fossil fuel import dependency), an industrial policy story (maintaining the nuclear supply chain's competitiveness through export volume), and a geopolitical positioning story (providing a democratic, rules-based alternative to Russian and Chinese nuclear in markets that have chosen to exit both). It is, in every dimension, a model for how a mid-size democracy can play the energy transition game.

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All data from KHNP, IAEA, World Nuclear Association, MOTIE, and cited news sources.